

Thesis

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September 7, 2026

1 Introduction

Dark matter has evaded detection efforts since it's prediction, in the eighties [RFT80; Rub; Shi21], indicating the potential need for an alternative. In this paper, I explore the effect of variable mass (variable higgs field vacuum expectation) on galactic rotation curves. This study is based off findings of universal acceleration, and spectra shifts due to spatially dependant mass from [Shi21].

References

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- [Shi21] Yuan Shi. *Force, Metric, or Mass: Disambiguating Causes of Uniform Gravity*. Sept. 3, 2021. DOI: 10.48550/arXiv.1908.02159. arXiv: 1908.02159 [gr-qc]. URL: <http://arxiv.org/abs/1908.02159> (visited on 04/14/2026). Pre-published.
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